

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1.	(a)			Variable resistor added in series with lamp (1) accept any size box with an arrow through it Voltmeter added in parallel with lamp (1)	2			2		2
	(b)	(i)		When current is 0.5 [A] the voltage is 0.9 ± 0.1 [V] (1) When current is 1 [A] the voltage is 2.4 ± 0.1 [V] (1) Triple would give 2.7 [V] or this is not triple (1) so not true To award 3 marks conclusion must be present Alternative 1: When the current doubles from 0.5 to 1 [A] (1) The voltage changes from 0.9 to 2.4 [V] (1) which is 2.7 times bigger or this is not triple (1) so not true To award 3 marks conclusion must be present Alternative 2: When current is 1 [A] the voltage is 2.4 ± 0.1 [V] (1) When current is 2 [A] the voltage is 8.4 ± 0.1 [V] (1) Triple would give 7.2 [V] or which is 3.5 times bigger or this is not triple (1) so not true To award 3 marks conclusion must be present Alternative 3: When the current doubles from 1 to 2 [A] (1) The voltage changes from 2.4 to 8.4 [V] (1) which is 3.5 times bigger or this is not triple (1) so not true To award 3 marks conclusion must be present			3	3		3